

17 Ans: A

For the object to remain in contact with the platform throughout the motion, its acceleration must not be greater than 9.81 m s^{-2} .

$$a_{\text{max}} = \omega^2 x_0$$

$$x_0 = \frac{a_{\text{max}}}{\omega^2} = \frac{a_{\text{max}}}{(2\pi f)^2} = \frac{9.81}{(2\pi(1.5))^2} = 0.11 \text{ m}$$

18 Ans: B